

Project Idea 1

1. Project Title: Smart Health Monitoring and Assistance Stick for Elderly and Visually Impaired

2. Purpose

Objective: To design and build an Arduino-based smart stick that continuously monitors an elderly person's body temperature and heart rate, while also acting as a blind-assistance stick by detecting nearby obstacles and alerting the user through sound or vibration.

Scope: The system can be used by:

- Elderly people who need regular health monitoring.
- Visually impaired persons for navigation and obstacle detection.
- Healthcare and caretakers, allowing easy observation of vital data in real-time.

This prototype can be adapted into a wearable health monitor, a smart cane, or even integrated into IoT health systems.

Significance: Enhances safety and independence of elderly and visually impaired users.

- Provides real-time health feedback, reducing emergency risks.
- Demonstrates how IoT and embedded systems can improve human well-being.
- Low-cost and easy-to-build, suitable for community healthcare initiatives or student innovation projects.

3. Components

- Microcontroller: Arduino Uno / Nano
- Sensors:
 - 1. LM35 or DS18B20 (Temperature Sensor)
 - 2. Pulse Sensor (KY-039 or MAX30102)
 - 3. Ultrasonic Sensor (HC-SR04)
- Actuators:
 - 1. Buzzer
 - 2. Vibration Motor
- Body/Chassis: Walking Stick or Cane
- Additional Components: 16x2 I2C LCD / OLED Display Breadboard & Jumper Wires
9V Battery or Li-ion Power Bank Enclosure Box

4. Cost Breakdown (Approx.)

Components	Unit Cost	Total
Arduino Uno / Nano	700	700

Ultrasonic Sensor (HC-SR04)	200	200
LM35 / DS18B20 Temp Sensor	150	150
Pulse Sensor	250	250
Buzzer	50	50
Vibration Motor	100	100
16x2 I2C LCD Display	350	350
9V Battery + Connector	120	120
Stick / Cane Body	200	200
Wires, Breadboard, Casing, etc.	180	180
Total Estimated Cost		≈ 2,300 BDT

5. Functionality Breakdown

- **Functionality 1: Health Monitoring (Temperature & Heart Rate)**
 - **Overview:** Continuously measures the user's body temperature and heart rate using LM35 and pulse sensor, and displays the readings on the LCD.
 - **Working Procedure:**
 - The LM35 sensor converts body temperature into a corresponding analog voltage, read by Arduino's analog pin.
 - The pulse sensor detects blood flow pulses through the fingertip, processed by Arduino to calculate BPM (Beats Per Minute).
 - The readings are updated on the LCD display every second.
 - If the temperature or BPM exceeds preset limits, the buzzer sounds to alert the user.
 - Power is supplied from a 9V battery, regulated through the Arduino board.
- **Functionality 2: Obstacle Detection for Blind Assistance**
 - **Overview:** Detects objects ahead of the user and provides an alert through buzzer or vibration motor.
 - **Working Procedure:**
 - The HC-SR04 ultrasonic sensor emits ultrasonic waves and calculates distance based on echo time.

- When the detected distance is less than a safe threshold (e.g., 60 cm), the Arduino activates the buzzer and vibration motor.
 - The closer the object, the stronger or faster the vibration can be programmed.
 - Power is shared with the main Arduino circuit for compactness.
- **Functionality 3: Output Alerts and Display**
 - **Overview:** Displays sensor readings and provides immediate feedback to the user.
 - **Working Procedure:**
 - The LCD display (or OLED) shows live temperature, BPM, and distance values.
 - The buzzer/vibration motor serves as alert outputs, controlled via Arduino digital pins.
 - The power system ensures the circuit runs efficiently with low current consumption.
 - The stick remains portable, powered by a 9V battery or rechargeable pack.

Project Idea 2

1. Project Title: Solar-assisted Autonomous Micro-Irrigation Rover (for home / community vegetable beds)

2. Purpose

Objective: To design and build an autonomous irrigation rover that patrols small vegetable beds or rooftop gardens, detects soil dryness, and dispenses water only where needed. The system aims to conserve water, reduce human effort, and operate sustainably using solar power for battery charging.

Scope: The robot will autonomously navigate along plant rows, measure soil moisture, and irrigate dry areas using a controlled water pump. It will make decisions based on light intensity (to avoid watering during strong sunlight) and battery level (to ensure sufficient energy before operation). The rover can be applied to home gardens, community farming plots, and rooftop gardens commonly found in Bangladesh.

Significance:

- Addresses water scarcity and wastage in urban and rural agricultural practices.
- Reduces manual labor for small farmers and gardeners.
- Promotes sustainable agriculture through solar energy utilization and intelligent resource use.
- Scalable and low-cost for widespread implementation in Bangladesh's agricultural communities.

3. Components

- Microcontroller: Arduino Uno (or Arduino Nano)
- Sensors:
 1. Capacitive Soil Moisture Sensor (×2)
 2. Ultrasonic Sensor (HC-SR04) for obstacle detection
 3. Light Sensor (LDR) for sunlight detection
- Actuators:
 1. DC Gear Motors (×2) for rover movement
 2. 12V Mini Water Pump (for irrigation)
- Body/Chassis: 2-wheel or 4-wheel rover chassis with caster wheel
- Additional Components:
 - L298N Motor Driver
 - IRLZ44N MOSFET (for pump switching)
 - 12V rechargeable battery (7Ah)
 - 10W–20W solar panel with charge controller
 - Breadboard, jumper wires, tubing for water delivery, water container, resistors (10kΩ for LDR), diodes, fuse

4. Cost Breakdown (Approx.)

Components	Unit Cost	Total
Arduino Uno	450	450
L298N Motor Driver	300	300
DC Gear Motors (×2)	250	500
Chassis with Wheels	600	600
12V Mini Water Pump	400	400
Capacitive Soil Moisture Sensors (×2)	200	400
Ultrasonic Sensor (HC-SR04)	150	150
LDR + 10kΩ resistor	30	30
IRLZ44N MOSFET	80	80
Solar Panel (10–20W)	1200	1200
12V 7Ah Battery	1000	1000
Solar Charge Controller	500	500
Breadboard + Wires + Tubing + Misc.	500	500
Estimated Total Cost		≈ 6,060 BDT

5. Functionality Breakdown

- **Functionality 1:** Soil Moisture Detection and Targeted Irrigation
- - **Overview:**

The rover autonomously measures soil moisture using sensors embedded beneath the chassis. When the soil moisture level drops below a certain threshold, it activates the water pump to irrigate that area precisely.
 - **Working Procedure:**
 - Sensors: Capacitive moisture sensors continuously monitor soil humidity through analog readings.
 - Logic: Arduino compares readings to a pre-calibrated dry/wet threshold.

- Actuator Response: If soil is dry → Arduino triggers MOSFET → powers water pump for a fixed duration (e.g., 5–10 seconds).
 - Control Logic:
 - IF soil_moisture < threshold AND light_intensity < sunlight_threshold:
 - start_pump()
 - delay(T)
 - stop_pump()
 - ELSE
 - continue_navigation()
 - Power Management: Pump and motors draw from the 12V battery; MOSFET ensures safe switching.

- **Functionality 2: Navigation and Obstacle Avoidance**
 - **Overview:**

The rover autonomously moves in a systematic path across the garden bed. It uses an ultrasonic sensor to detect obstacles and avoid collisions while patrolling.
 - **Working Procedure:**
 - Sensors: HC-SR04 measures distance in front of the robot.
 - Logic: Arduino checks for obstacles every few seconds while moving forward.
 - Actuator Response:
 - If obstacle < 25 cm → stop motors → turn left/right → continue.
 - If clear → move forward at set speed.
 - Motor Control:
 - L298N driver controlled by 4 digital pins for direction (forward, reverse, left, right).
 - Speed regulated using PWM signals from Arduino.
 - Power Management: 12V supply powers both motors through L298N.

- **Functionality 3: Solar-Powered Energy and Environmental Awareness**
 - **Overview:**

The system uses an LDR to measure ambient light intensity and make intelligent irrigation decisions. It avoids watering under direct sunlight and utilizes solar charging for sustainability.
 - **Working Procedure:**
 - Sensors: LDR connected via voltage divider to analog pin A1.
 - Logic:
 - If sunlight intensity is high (bright day) → delay watering to reduce evaporation.
 - If low (morning/evening) → allow irrigation.
 - Solar System Integration:

1. The solar panel charges a 12V battery through the charge controller.
 2. Battery voltage is periodically read via analog divider to estimate capacity.
 3. When battery voltage $< 11V \rightarrow$ rover halts watering and enters “low-power” mode until recharged.
- Control Logic:
IF light_value $<$ LIGHT_THRESHOLD AND battery_voltage $>$ 11.0V:
 irrigation_enabled = true
ELSE
 irrigation_enabled = false

Project Idea 3

1. Project Title: Canal / Low-Lying Area Flood Early Warning Rover with GSM Alert

2. Purpose

Objective: To develop an autonomous flood-monitoring rover that patrols along canal or embankment paths, measures water level and rain intensity, and issues local (buzzer/LED) and remote (SMS via GSM) flood warnings when rising water levels exceed safety thresholds.

Scope:

1. Designed for low-lying and flood-prone regions of Bangladesh where local monitoring systems are scarce.
2. Can be deployed along canals, small rivers, and drainage systems to provide early flood alerts.
3. The rover operates autonomously, measuring water levels at intervals and sending SMS alerts to local authorities or community members.
4. The system can be expanded with IoT cloud connectivity or solar charging for long-term sustainability.

Significance:

1. Provides low-cost, decentralized flood monitoring in rural and semi-urban regions.
2. Uses readily available components and GSM networks, making it practical in Bangladesh.
3. Helps local communities take early action before water levels become dangerous.
4. Encourages sustainable, technology-driven disaster preparedness through automation and local innovation.

3. Components

- Microcontroller: Arduino Uno / Nano
- Sensors: - Ultrasonic Sensor (HC-SR04 / JSN-SR04T waterproof) – for measuring distance to water surface.
- Rain Sensor – detects rainfall intensity.
- (Optional) DHT22 Temperature & Humidity Sensor – environmental data.
- (Optional) Turbidity/Conductivity Sensor – water quality indication.

- Actuators:DC Motors ×2 (for mobility), L298N Motor Driver, Buzzer (active)
- Body/Chassis: 4-wheel or Tracked Rover Base
- Additional Components: Logic level shifter, Waterproof casing, Fuse and switch, Float switch (optional redundancy)

4. Cost Breakdown (Approx.):

Components	Unit Cost	Total
Arduino Uno / Nano	700	700
L298N Motor Driver	300	300
DC Motors (×2)	400	800
Rover Chassis (4-wheel / tracked)	800	800
Ultrasonic Sensor (JSN-SR04T Waterproof)	350	350
Rain Sensor	150	150
SIM800L GSM Module	600	600
Buzzer	80	80
LED Strobe	100	100
12V Battery Pack + Regulator	600	600
Misc. (wires, casing, waterproofing, connectors, switch)	300	300
Total Estimated Cost		≈ 4,780 TK

5. Functionality Breakdown

- **Functionality 1: Water Level & Rain Monitoring**
 - **Overview:** Measures real-time water height in canals or low-lying areas and detects rainfall intensity to assess potential flood conditions.
 - **Working Procedure:**
 1. The ultrasonic sensor faces downward toward the water surface; distance readings are converted to water level by calibration.
 2. The rain sensor measures moisture intensity on its plate.

3. Arduino processes both readings and compares them to threshold values (early_warning, critical_level).
4. If the water level rises beyond the early threshold, the buzzer beeps and LED flashes as a local alert.
5. Data can also be logged or transmitted later if connected to IoT/cloud.
6. Power is drawn from a 12V battery, with voltage regulators supplying 5V for Arduino and 4V for SIM800L.

- **Functionality 2: GSM-Based Flood Alert System**

- **Overview:** Sends SMS alerts to registered phone numbers when water level crosses early or critical warning limits.
- **Working Procedure:**
 1. Arduino uses the SIM800L GSM module for SMS transmission.
 2. Upon reaching a defined threshold, Arduino constructs a text alert such as: "Warning: Water level rising rapidly! Level: 85cm. Take precautions."
 3. The message is sent to multiple phone numbers (e.g., local ward office, disaster cell).
 4. SMS function uses AT commands via SoftwareSerial interface (pins D10, D11).
 5. To prevent spam, alerts are debounced — sent only once per event until reset.
 6. The GSM module requires a stable 3.7–4.2V supply capable of 2A peak current.
 7. Power management uses a LiPo battery or regulated output from the main supply.

- **Functionality 3: Autonomous Rover Patrolling & Local Alert**

- **Overview:** The rover patrols embankments or canal edges automatically and reacts to environmental changes.
- **Working Procedure:**
 1. Two DC motors controlled via L298N motor driver allow forward, backward, and turning motion.
 2. The rover follows a predefined route or moves until an obstacle is detected.
 3. At each stop or waypoint, it takes sensor readings (water level, rain intensity).
 4. If an obstacle or deep water area is encountered, the rover changes direction automatically (basic obstacle avoidance).
 5. On flood warning conditions, the rover activates LED strobe and buzzer, providing immediate local alert.

6. If battery voltage drops below threshold, the rover returns to base station or stops for safety.